

DRAFT — Petroleum Sector Framework: Design Basis and Military Action Extension

Design Basis and Military Action Extension

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Preface

The petroleum sector occupies a foundational position in modern industrial civilization. Crude oil, refined into gasoline, diesel, jet fuel, heating oil, and the feedstocks for petrochemical production, powers transportation, military logistics, agricultural supply chains, and the manufacturing base on which every other sector depends. Yet the sector that underlies all others operates without a formal Design Basis — without a documented, methodologically rigorous statement of the bounding events its facilities must be designed to survive. Regulation addresses pipeline corrosion. Environmental law addresses spill response. Safety regulations address process hazards. But no unified framework asks the foundational question: what is the worst credible event this facility must withstand, and what does adequate protection against that event require?

This document fills that gap and immediately extends it. The DBA-OIL sector framework establishes a Design Basis for the petroleum sector — a structured methodology for identifying bounding events, defining consequence envelopes, and specifying the system response standards that constitute adequate protection. It then extends that basis to the military action threat tier, asking what happens when the bounding event is not an equipment failure, a natural hazard, or an operator error, but rather a deliberate, sophisticated, state-directed attack against petroleum infrastructure.

The evidentiary record is unambiguous. Russian airstrikes on Ukrainian petroleum refining and storage infrastructure in 2022 through 2024 systematically destroyed refinery capacity, disrupted fuel supply for civilian and military consumers, and forced the Ukrainian economy to absorb the cascading consequences of domestic refinery loss — increased import dependency, fuel price escalation, and the diversion of scarce foreign exchange to petroleum product procurement. The Islamic Revolutionary Guard Corps has targeted Saudi Aramco facilities with cruise missiles and UAS. The Houthis' maritime campaign has threatened Gulf tanker routes. The design basis is drawn from events that have already occurred.

The DBA-OIL framework is a companion to the DBA-GAS framework, which addresses the natural gas sector under the same analytical architecture. Together, they provide the design basis for the petroleum and gas sector that federal regulators, industry operators, and allied defense planners need to assess, prepare for, and mitigate the consequences of military action against energy infrastructure.

1. Introduction

1.1 Purpose

This document establishes the Design Basis and Design Basis Accident — Military Action (DBA-MA) framework for the petroleum sector. It serves two related purposes.

First, it introduces the Design Basis concept to the petroleum sector and establishes the methodology for identifying and characterizing bounding events — the events that petroleum sector facilities must be designed to withstand. The non-adversarial Design Basis defined in Section 4 establishes the baseline resilience posture against which all subsequent analysis is measured.

Second, it extends that Design Basis to the military-action threat tier. Section 5 introduces the four characteristics of military action that distinguish it from non-adversarial bounding events and explains why these characteristics require an extended analytical framework. The DBA-MA extension is a natural and necessary development of the Design Basis, not a separate methodology imposed upon it.

1.2 Scope and Applicability

This document governs the DBA-OIL series. It applies to petroleum sector infrastructure, including upstream crude oil production (onshore and offshore), petroleum refining and processing, crude oil and product pipelines, marine terminals and bulk liquid storage, and associated control systems. The DBA-OIL series primarily addresses the U.S. petroleum sector, with reference to allied and international contexts in which the consequences of military action extend beyond domestic boundaries.

Natural gas production, processing, and transmission are addressed in the DBA-GAS series. Where petroleum and natural gas infrastructure intersect — offshore platforms producing both liquids and gas, processing facilities handling mixed streams, or pipeline systems carrying natural gas liquids — this document addresses the consequence chain as it originates in the petroleum sector, without displacing the DBA-GAS analysis of the gas-sector consequences of the same event.

1.3 Relationship to Governing Documents

This document is issued under the DBA-MA Parent Framework (IAEA L1 v1.4). The parent framework establishes the governing threat taxonomy, consequence analysis methodology, and framework architecture. This sector framework inherits from the parent and adapts those foundations to the petroleum sector's specific characteristics.

Facility-class documents, use cases, and policy companions in the DBA-OIL series are issued under this sector framework. They inherit the threat taxonomy, bounding methodology, and analytical structure established here without repeating it. DBA-OIL-FC1 (Petroleum Refining) is currently issued under this framework. Additional facility-class documents for offshore production, terminals and storage, and transport infrastructure are anticipated as the series matures.

1.4 How to Use This Document

A Design Basis answers a specific engineering question: what is the worst credible event this facility or system must be built to survive? Not the worst imaginable event — the worst credible one. Once the envelope is defined, the engineering question becomes tractable: which facility design, operating procedures, and recovery architecture constitute adequate protection against the bounding event?

The Design Basis Accident — Military Action extension asks the harder version: what is the worst credible event when the initiating condition is deliberate military action? The answer is harder because an adversary adapts, chooses targets, and times attacks to maximize consequences. The analysis in this document provides the tools to answer it.

Readers primarily interested in the military-action analysis may proceed directly to Section 5 after reading Section 3. Readers responsible for facility-class document development should read the full document, as the sector framework establishes the analytical inheritance that all facility-class documents carry.

2. Sector Profile

2.1 Sector Structure

The petroleum sector is organized into four operational layers. Each layer has distinct physical characteristics, regulatory oversight, and consequence profiles under both non-adversarial and military-action conditions.

2.1.1 Upstream: Crude Oil Production

Upstream infrastructure includes producing wells, wellheads, gathering pipelines, and field separation equipment for both onshore and offshore production. The United States is the world's largest crude oil producer, with major producing basins concentrated in the Permian Basin (Texas and New Mexico), the Gulf of Mexico outer continental shelf, the Bakken formation (North Dakota), and the Eagle Ford and Anadarko basins. Offshore Gulf of Mexico production accounts for approximately fifteen percent of total U.S. crude production and is concentrated on a relatively small number of major platforms and deepwater installations.

Onshore production is geographically dispersed, providing inherent resilience against single-node disruptions. Offshore production presents a different profile: a relatively small number of platforms and pipeline systems account for a large share of offshore production volume, and the remote, maritime operating environment imposes unique access and recovery constraints under military action conditions.

2.1.2 Midstream: Refining and Processing

Petroleum refining is the operationally critical and most consequential layer of the petroleum sector. The United States has approximately 130 operating refineries with a combined crude distillation capacity of approximately 18 million barrels per day. Refinery capacity is geographically concentrated: the Gulf Coast (PADD 3) accounts for approximately 45 percent of total U.S. refining capacity, creating a regional concentration vulnerability analogous to the Gulf Coast's role in natural gas production and processing.

Refineries are among the most process-complex facilities in the energy sector. They operate continuously, require months of planning for scheduled maintenance shutdowns, and have recovery timelines after unplanned outages that depend on the nature and extent of the damage. A refinery fire or explosion that damages process units — distillation columns, fluid catalytic crackers, hydrotreaters, alkylation units — can require six to eighteen months for full restoration, and some process units have replacement lead times that extend beyond the repair timeline. The multi-product consequences of a refinery outage — the simultaneous loss of gasoline, diesel, jet fuel, heating oil, and petrochemical feedstock production — distinguish refinery attacks from single-product infrastructure disruptions.

2.1.3 Midstream: Pipelines and Transport

Crude oil trunk pipelines move crude from producing regions to refineries. Product pipelines distribute refined products from refineries to distribution terminals and end markets. The Colonial Pipeline system — the largest refined product pipeline in the United States — carries approximately 100 million gallons per day of gasoline, diesel, and jet fuel from Gulf Coast refineries to markets along the Eastern Seaboard. The 2021 ransomware attack on Colonial demonstrated that a non-state cyber actor could produce regional fuel supply disruptions with national visibility; the DBA-MA design basis addresses the state-actor version of the same attack with kinetic amplification.

Marine transport moves crude oil from producing regions and import terminals to refineries, and moves refined products from refinery docks to coastal and export markets. Jones Act tankers serve domestic routes; international tankers serve import and export flows. The concentration of crude import terminals and refinery docks in the Gulf Coast and East Coast creates maritime approach corridors whose disruption affects the land-side supply chain.

2.1.4 Downstream: Terminals, Storage, and Distribution

Petroleum product terminals receive product from pipelines and marine vessels, store it in bulk liquid tanks, and distribute it by truck, rail, or local pipeline to retail and commercial customers. Terminal tank farms are geographically distributed but concentrated at ports and major pipeline termini. The consequence of terminal disruption is measured in days to weeks of regional product supply impact, shorter than the consequence of refinery disruption but more immediately visible to consumers.

The Strategic Petroleum Reserve (SPR) — approximately 350 million barrels of crude oil stored in salt caverns on the Gulf Coast — is the primary federal emergency supply mechanism for crude oil shortages. The SPR has no direct equivalent in the natural gas sector. Its existence as a response tool affects the consequence analysis for crude supply disruption events. However, its geographic concentration on the Gulf Coast means it shares some of the same vulnerability exposure as the refinery capacity it is intended to support.

2.2 Regulatory Authority

The petroleum sector operates under a fragmented regulatory structure with jurisdiction divided among safety, environmental, economic, and security authorities. PHMSA holds pipeline safety jurisdiction under 49 CFR Parts 192 and 195. EPA holds environmental response jurisdiction under the Clean Water Act, the Oil Pollution Act, and RCRA; the OPA 90 framework requires facility response plans for worst-case discharge scenarios, but does not address military-action threat conditions. The Bureau of Safety and Environmental Enforcement (BSEE) holds offshore production safety jurisdiction. FERC holds limited economic jurisdiction over oil pipelines under the Interstate Commerce Act. TSA holds surface transportation security authority and has issued mandatory cybersecurity directives following the Colonial Pipeline incident; physical security requirements for pipelines remain voluntary. CISA provides sector risk management support with advisory authority only.

The same mandatory physical security gap that characterizes the natural gas sector applies equally to the petroleum sector. No mandatory physical security standard governs petroleum refinery, terminal, or pipeline facilities. The deterrence and resilience implications of this gap are addressed in the DBA-OIL policy companion.

2.3 Strategic Significance

The petroleum sector's strategic significance derives from three interdependency arguments that elevate a supply disruption from an energy event to a national security event.

Military logistics dependency. Petroleum products — jet fuel for aircraft, diesel for ground vehicles and naval vessels, and marine fuel for the logistics fleet — are the operational fuel of military power projection. A sustained disruption of domestic refining capacity that degrades jet fuel and diesel production directly constrains the military operational tempo. An adversary that simultaneously threatens the alliance militarily and degrades the domestic refining capacity that supports military logistics has created a compound problem: the military response is constrained by the very attack that necessitates it.

Transportation and commerce dependency. The U.S. economy moves on petroleum products. Trucking, aviation, and rail freight depend on diesel and jet fuel from domestic refineries. A sustained refinery outage on the Gulf Coast — the region that produces nearly half of U.S.

refined products — produces supply shortfalls that cascade through freight costs, agricultural supply chains, and manufacturing logistics within days.

Allied energy security dependency. U.S. petroleum exports — crude oil, refined products, and natural gas liquids — are a component of the allied energy security architecture. A sustained disruption of Gulf Coast refining and export terminal capacity affects allied nations that depend on U.S. product exports as a marginal source of supply. The alliance-level consequence of a major U.S. refinery disruption is less acute than that of an equivalent disruption to LNG export capacity. Still, it is not negligible for allied nations with limited domestic refining capacity.

3. Design Basis: Petroleum Sector

3.1 The Concept and Its Origins

The Design Basis concept was formalized in nuclear safety engineering in the 1960s and 1970s as the foundation of reactor safety analysis. The nuclear industry faced a specific analytical challenge: reactors could fail in many ways, but design and licensing resources were finite. A methodology was needed to identify the bounding cases — the worst credible events that safety systems must be designed to withstand — and to distinguish them from less severe events that existing safety margins would absorb without specific design accommodations.

The result was the Design Basis Accident: a postulated accident that represents the most severe conditions that safety systems must be capable of mitigating. The DBA is not the worst imaginable accident. It is the worst accident that can credibly result from defined equipment failures, operator errors, or natural hazards, given the facility's design and operating envelope. Once the DBA is defined, the safety system design requirement follows directly.

The concept's extension to the petroleum sector is a direct application of the same analytical logic. The petroleum sector presents the same foundational challenge: a complex system with many possible failure modes, finite design and investment resources, and a need to identify the bounding cases that drive hardening and resilience requirements.

3.2 Application to the Petroleum Sector

Applying the Design Basis methodology to the petroleum sector requires adapting the nuclear framework to a fundamentally different system architecture. Nuclear facilities are single-point systems with defined safety systems and consequence chains that proceed outward from a single facility. The petroleum sector is a networked system of geographically distributed but functionally interdependent facilities.

The Design Basis for the petroleum sector is defined at two levels. At the facility level, the Design Basis addresses the bounding events for individual asset classes: what is the worst credible event a petroleum refinery must be designed to survive? At the system level, it addresses

the bounding events for the petroleum supply network as an integrated system: what combination of facility-level events produces the worst credible systemic supply disruption?

A key distinction from the natural gas sector: the petroleum sector's cascade consequences are primarily economic and logistical rather than immediate life-safety. A refinery outage does not immediately threaten lives the way a heating gas supply disruption during a winter storm does. The consequence timeline is longer, the immediate life-safety consequence is lower, and the recovery mechanisms — including the SPR and international crude markets — are more developed. These differences shape the adequacy standards in Section 4.2 and the bounding events in Section 4.

The Design Basis serves a second function beyond establishing engineering requirements: it is a gap-finding instrument for policy, authority, and institutional coordination. Defining precisely what the sector must withstand makes visible the delta between that requirement and what existing regulatory frameworks actually compel. Every adequacy standard without a compliance mechanism, and every cross-sector coordination requirement without a pre-established protocol, is a gap that the Design Basis exposes—section 10 catalogs what this document reveals.

4. Design Basis Accidents: Non-Adversarial Envelope

This section constitutes the DBA layer — the second stage of the DB → DBA → DBA-MA sequence. The Design Basis in Section 3 defines the petroleum sector, explains why it is critical, and outlines the performance envelope. This section derives the specific postulated accident categories from that envelope and establishes the methodology and adequacy standards for analyzing each category. The DBA-MA extension in Section 5 then applies the four military action characteristics to each category. Non-adversarial bounding events fall into four categories: process safety events, natural hazard events, operational events, and cascading failure events. The list is illustrative, not exhaustive; the framework will be updated as operational experience and analysis surface new candidates.

4.1 Bounding Event Methodology

A bounding event is identified through a four-step process of hazard identification, consequence analysis, credibility assessment, and bounding selection. This methodology is established in the DBA-GAS Sector Framework and inherited here without repetition. The petroleum sector's application of the methodology yields different bounding events than the gas sector's because the hazard profile, consequence chains, and recovery mechanisms differ.

The bounding events identified in this document and in facility-class documents are not exhaustive. They represent the current best assessment of the most severe credible events for each facility class and system layer. As the DBA-OIL series matures and facility-class documents develop deeper analyses, additional bounding events will be identified and

incorporated. The methodology is the stable foundation; the specific bounding events it produces are subject to revision.

4.2 Design Basis Adequacy Standards

Adequate system response to the bounding event is defined against three criteria for the petroleum sector.

Supply continuity for priority products. The system must maintain the supply of priority petroleum products — military fuel grades, diesel for critical freight and emergency services, and heating oil for residential customers in regions dependent on petroleum heat — within defined supply-stress limits following the bounding event. The SPR provides the primary emergency crude supply mechanism; adequacy standards for refinery resilience must account for SPR drawdown as a bridge supply rather than a substitute for refinery capacity.

Recovery capability. The system must possess the redundancy, spare equipment inventory, and logistics capability needed to restore normal operations within a defined recovery window. Recovery capability requirements drive investment in spare process-unit components, firefighting infrastructure, and mutual-aid arrangements that are currently absent from any mandatory framework.

Environmental consequence limitation. Petroleum facility failures produce environmental consequences — crude oil spills, product releases, and hazardous material releases — that extend the chain of consequences beyond the supply disruption. Adequacy standards must include limits on environmental consequences, particularly for offshore production and marine terminal scenarios, where damage to the aquatic environment compounds the supply and economic consequences of the initiating event.

4.3 Process Safety Events

4.3.1 Major Refinery Fire and Explosion

Initiating event: Uncontrolled hydrocarbon release at a major process unit — distillation column, fluid catalytic cracker, hydrotreater, or alkylation unit — resulting in fire or explosion that damages the process unit and potentially adjacent units. The 2005 Texas City refinery explosion is the reference event: a raffinate splitter tower overfilled and released hydrocarbon vapor, which ignited and killed fifteen workers. More consequential for the Design Basis is the Amuay refinery explosion in Venezuela (2012), where a gas leak at a crude distillation unit ignited and caused an explosion that killed 48 people and destroyed a significant portion of the facility.

Consequence envelope: Physical destruction of one or more process units. Loss of production across all product streams from the affected units. Recovery timeline of six to eighteen months for major unit reconstruction, with variation depending on the specific unit type, the extent of damage, and the availability of replacement equipment. Refinery fires frequently cascade — the

initial event damages adjacent piping and equipment, fire-suppression water and foam runoff create environmental contamination, and the facility may be unable to restart any units until the full extent of damage is assessed and remediated.

Adequacy standard: Process safety management systems (PSM) under OSHA 29 CFR 1910.119 establish baseline requirements for process hazard analysis and mechanical integrity. For Design Basis purposes, the adequacy standard requires that catastrophic unit loss does not simultaneously disable the facility's fire suppression capability, that secondary containment prevents runoff contamination from exceeding defined environmental limits, and that the facility's emergency response plan addresses the multi-unit escalation scenario.

4.3.2 Large Crude Oil Spill — Pipeline

Initiating event: Rupture of a major crude oil trunk pipeline from corrosion, third-party damage, or geotechnical instability, resulting in a large-volume crude oil release to the surrounding environment. The 2010 Enbridge Line 6B rupture in Marshall, Michigan, is a reference event: a corroded pipeline segment failed, releasing approximately 843,000 gallons of diluted bitumen crude into a wetland and the Kalamazoo River, prompting a cleanup effort that took four years and cost approximately \$1 billion.

Consequence envelope: Crude oil release to soil and water. Environmental remediation costs and timelines that significantly exceed the costs of physical pipeline repair. Supply curtailment to refineries served by the affected pipeline during the isolation and repair period. Pipeline restart requires PHMSA approval following hydrostatic testing and remediation assessment, adding weeks to months beyond the physical repair timeline. The bounding case involves a crossing of navigable waters or an environmentally sensitive wetland, where remediation requirements are most stringent.

Adequacy standard: Block valve spacing and actuation capability must limit release volume. Secondary containment and spill response equipment must be pre-positioned at high-consequence locations, as defined under PHMSA's high-consequence area framework. Alternative crude supply routing or refinery crude flexibility must sustain refinery operations during the repair window.

4.3.3 Offshore Platform Blowout

Initiating event: Loss of well control at an offshore production platform, resulting in an uncontrolled release of crude oil and associated gas. The Deepwater Horizon blowout (2010) is the reference event: a blowout preventer failure during well completion operations caused an explosion that killed eleven workers, sank the drilling rig, and released approximately 3.19 million barrels of crude oil into the Gulf of Mexico over 87 days.

Consequence envelope: Immediate fatalities and injuries at the platform. Total production loss from the affected platform. Environmental damage across the affected marine area, with

shoreline impacts requiring multi-year remediation—regulatory response limiting operations across the affected basin. The Deepwater Horizon event resulted in a six-month drilling moratorium in the Gulf of Mexico, affecting production from other installations, independent of any physical damage to them. The bounding case accounts for this regulatory cascade as a consequence of the initiating event.

Adequacy standard: BSEE’s well control regulations require blowout preventer testing and well control plans that reduce the probability of uncontrolled release. For Design Basis purposes, the adequacy standard requires that offshore production facilities have surface and subsea well control capability sufficient to achieve well shut-in within a defined time limit, and that oil spill response organizations (OSROs) are pre-contracted and positioned for the worst-case discharge scenario.

4.4 Natural Hazard Events

4.4.1 Hurricane Disruption of Gulf Coast Refining

Initiating event: A major hurricane (Category 3 or above) making landfall on or traversing the Gulf Coast, causing simultaneous physical damage, flooding, and extended power outages across the concentrated Gulf Coast refinery cluster. Hurricanes Katrina and Rita (2005) are the reference events: those storms simultaneously shut down approximately 30% of U.S. refining capacity, causing spikes in refined product prices and supply shortfalls that affected markets nationwide.

Consequence envelope: Physical damage to refinery process units from wind and storm surge. Extended power outages are affecting refinery restart capability, independent of physical damage. Damage to road and rail infrastructure is limiting personnel access and feedstock delivery. The bounding case involves a direct strike on the Houston Ship Channel/Beaumont refinery cluster, which accounts for approximately 20% of total U.S. refining capacity. Recovery from major physical damage at multiple refineries simultaneously requires the same scarce specialty contractors, extending each facility’s recovery timeline beyond what it would be for an isolated event.

Adequacy standard: Refinery infrastructure in the Gulf Coast hurricane exposure zone must meet design standards for wind loading and storm surge appropriate to the design storm intensity. Emergency response plans must address simultaneous damage across multiple facilities and include pre-established mutual-aid arrangements for specialty-contractor resources. Product pipeline operators must have plans for emergency product supply routing from alternative refinery sources during Gulf Coast outage periods.

4.4.2 Seismic Event — West Coast Refineries

Initiating event: A significant seismic event (magnitude 6.5 or greater) affecting the West Coast refinery concentration in the Los Angeles Basin or San Francisco Bay Area. West Coast refineries supply the majority of gasoline, diesel, and jet fuel for California — the largest state fuel market in the country — and have limited pipeline connectivity to other regional supply sources. A major earthquake affecting multiple West Coast refineries simultaneously would trigger a regional fuel supply crisis with limited alternatives.

Consequence envelope: Physical damage to refinery process units from ground motion. Fire and explosion risk from process unit damage in a seismically active event. Loss of utility power affects the restart capability. Limited alternative supply: unlike the Gulf Coast, the West Coast has minimal pipeline interconnection to other refining regions, and marine supply from Gulf Coast refineries requires two to three weeks of transit time. The bounding case involves a direct rupture of the Hayward or Newport-Inglewood fault, affecting the Bay Area or Los Angeles Basin refinery clusters, respectively.

Adequacy standard: West Coast refinery infrastructure must meet seismic design standards appropriate to the facility's location on the applicable fault hazard map. Emergency response plans must address the extended supply curtailment scenario and include pre-arranged supply agreements for marine product imports during a regional refinery outage.

4.5 Operational Events

4.5.1 Control System Failure — Refinery

Initiating event: Failure or loss of availability of the distributed control system (DCS) governing a major refinery's process operations, requiring emergency shutdown of process units and transition to manual operations. A refinery DCS failure during normal operations can force a controlled shutdown that takes days to weeks to restart safely; an uncontrolled DCS failure during an upset condition can trigger emergency shutdowns across multiple process units simultaneously, with the risk of secondary fires or releases during the shutdown sequence.

Consequence envelope: Loss of refinery production during the DCS failure and restart period. Safety risk during the shutdown transition if the failure occurs during a process upset. DCS restoration and process restart require sequential unit restarts in the correct order, with safety checks at each stage; the minimum restart time after a controlled DCS shutdown is typically several days. In the bounding case, a DCS failure initiates a process upset in an active unit, resulting in a secondary fire or release.

Adequacy standard: Refinery control systems must have redundant DCS architecture and manual control capability for emergency shutdown of each process unit. The transition to manual operations must be exercised periodically. Recovery time objectives for DCS restoration must be defined and achievable within the facility's emergency response plan.

4.5.2 Major Terminal Fire

Initiating event: Ignition of a petroleum product storage tank at a major product terminal, resulting in a tank fire that threatens adjacent tanks and terminal infrastructure. Tank fires can result from lightning strikes, static discharge, maintenance errors, or equipment failure. A tank fire that escalates to involve multiple tanks — a boilover event, in which water accumulated at the bottom of a tank is superheated and erupts violently, carrying burning crude oil over the tank walls — represents the bounding case for terminal fire scenarios.

Consequence envelope: Destruction of the affected tank and potential adjacent tanks: product loss and environmental contamination from firefighting runoff. Terminal operations suspended during fire suppression and remediation. The bounding case involves a major crude storage terminal where a boilover event affects multiple large tanks, overwhelming on-site firefighting capability and requiring mutual aid from regional fire resources.

Adequacy standard: Terminal fire suppression systems must be sized for the tank diameter and product type in accordance with NFPA 11 and API standards. Impoundment capacity must contain the contents of the largest tank plus firefighting water without offsite release. Emergency response plans must include pre-arranged mutual aid for large-tank firefighting.

4.6 Cascading Failure Events

4.6.1 Refinery-to-Transport Cascade

Initiating event: An event that simultaneously disables a major refinery cluster and the product pipeline serving its primary market, eliminating both production and transport redundancy for an extended period.

Consequence envelope: The refinery-to-transport cascade is the highest-consequence non-adversarial event in the petroleum sector's Design Basis. The Colonial Pipeline 2021 ransomware incident approximated the transport component in isolation, resulting in regional fuel shortages along the Eastern Seaboard within days. The combined refinery-and-pipeline scenario compounds the supply impact and eliminates the routing flexibility that would contain the consequences of either event in isolation.

Adequacy standard: Regional product supply resilience must be assessed at the system level, not only the facility level. Pre-established emergency supply protocols — SPR drawdown for crude, product import surge, demand reduction measures — must be sized for the combined refinery-and-pipeline bounding case, not for either event in isolation.

5. The Military Action Extension

5.1 Why the Non-Adversarial Design Basis Is Insufficient

Each bounding event in Section 4 shares a characteristic that distinguishes it from the military-action threat tier: it is not chosen. Process upsets occur according to the probabilistic distribution of equipment failure modes. Hurricanes follow meteorological paths. The system designer can analyze historical frequencies and invest in protection against the events most likely to occur within a defined planning horizon. The adversary does not operate within this framework.

Military action against petroleum infrastructure introduces the four characteristics that require an extended analytical framework. These characteristics were identified in the DBA-MA parent framework in the nuclear context and apply without modification to the petroleum sector.

5.2 The Four Military Action Characteristics

5.2.1 Intentionality

The adversary chooses its targets. It studies the petroleum sector, identifies the facilities whose loss would have the greatest strategic consequence, assesses the defensive posture of those facilities, and selects the attack configuration most likely to achieve the desired effect. The Gulf Coast refinery concentration is not a vulnerability that an adversary discovers by accident — it is publicly documented, economically significant, and geographically concentrated in a way that makes coordinated attack planning straightforward. An adversary who has studied the sector knows that the Houston Ship Channel refinery cluster, a single coastal corridor spanning approximately twenty miles, accounts for a material fraction of U.S. refined product capacity for gasoline, diesel, and jet fuel.

Intentionality means the adversary will not attack in August when an attack timed to the winter heating oil season or the summer driving season produces a larger consumer-visible consequence. It means the adversary will target the fluid catalytic cracker whose replacement lead time is longest, not the least consequential unit whose spares are pre-positioned. And it means the adversary will attack the facility that the sector has not hardened, not the one whose physical security posture would complicate the operation.

5.2.2 Persistence

Military action does not affect the sector's recovery timeline. A refinery fire is extinguished; repairs begin; the facility restarts in months. Military action can be sustained across the recovery period. The contractor crews mobilizing to repair blast-damaged process units are operating in a threat environment. The supply chain delivering replacement compressors and heat exchangers is a target. The product pipeline that would route supply around the damaged refinery during its recovery is itself a potential target in the second phase of the campaign.

The persistence characteristic is particularly consequential for the petroleum sector because refinery recovery timelines are already measured in months. An adversary who can disrupt the

recovery — by threatening the site or interdicting the supply chain for replacement equipment — can extend the effective disruption well beyond the period of physical damage.

5.2.3 Precision

Modern state actors can execute precision attacks against specific facility components. In a petroleum refinery, precision targeting allows the adversary to select the highest-consequence unit: the fluid catalytic cracker, whose loss eliminates gasoline production; the alkylation unit, whose replacement requires 18 months; or the hydrogen plant, whose loss shuts down the entire hydroprocessing train. Precision also allows target selection across a multi-facility campaign: attacking the same unit type at multiple refineries simultaneously eliminates the mutual-aid and routing flexibility that would contain the consequence of a single-facility event.

The Ukrainian experience is directly applicable: Russian strikes on Ukrainian petroleum refining infrastructure from 2022 through 2024 demonstrated that an adversary with precision-strike capability will target the specific process equipment whose loss maximizes disruption duration, rather than merely the most physically accessible target.

5.2.4 Denial

Military action can deny site access. The firefighting crews that would contain a process fire, the inspection teams that would assess damage and authorize restart, and the contractor crews that would execute repairs cannot access a facility under active military threat. The local sovereignty principle — that protective systems must operate passively, without external power, communications, or operator action — applies to the petroleum sector as it does to nuclear facilities.

Denial has a specific amplification mechanism in the petroleum sector: process upsets that are normally contained by prompt operator intervention can escalate to catastrophic events if operators are unable to access the facility or execute manual interventions. A refinery fire that the facility's own fire brigade would normally contain within hours can escalate to catastrophic unit loss if the brigade cannot approach the fire under military-action-threat conditions.

5.3 The Extended Design Basis: DBA-MA

The DBA-MA extension builds on the non-adversarial Design Basis rather than replacing it. Each non-adversarial bounding event has a military-action analog: the same initiating condition, deliberately caused, precisely targeted, strategically timed, and executed to defeat the recovery architecture.

The DBA-MA bounding event is in every case more severe than its non-adversarial analog. The adversary selects the highest-consequence configuration; the non-adversarial event occurs according to probability. The adversary times the attack for maximum effect; the non-adversarial event is timing-indifferent. The adversary coordinates simultaneous events that defeat single-

scenario resilience mechanisms; the non-adversarial events are independent. The adversary denies the access on which recovery depends; the non-adversarial recovery framework assumes unimpeded access.

The design-basis scenarios in DBA-OIL-FC1 instantiate the DBA-MA bounding event for the petroleum-refining facility class. Subsequent facility-class documents will instantiate it for offshore production, terminals and storage, and transport infrastructure.

5.4 The Command-Broker Architecture and the Bright Line

The command-broker architecture is the Design Basis requirement for all ML and Gen AI components whose outputs can affect physical process parameters in petroleum sector control systems. It establishes an architectural boundary -- the Bright Line -- between the advisory layer and the physical command layer. Gen AI may not cross the Bright Line autonomously. The boundary is enforced by architecture, not by procedure or operator discipline.

The canonical cross-sector instrument governing this architecture is the Command Broker Framework, Cross-Sector (CB-Framework_Cross_Sector v0.1), published by Eclectic Technologies. That document establishes the broker concept, the Bright Line principle, the three-tier criticality framework, and the distinction between the Command Broker and the Influencer Broker. It is the authoritative source for all requirements that this section implements in the petroleum sector context. The CB-Framework derives its three-tier criticality framework from the ISAGCA Supplier Ad-hoc Working Group's analysis of ML and Gen AI risk in critical infrastructure industrial control systems, published as "A Study of AI Risks in Critical Infrastructure -- Vendor Agnostic."

The Three-Tier Criticality Framework. The CB-Framework establishes three tiers based on consequence severity. At the highest criticality tier -- applications whose malfunction could produce the Design Basis consequence thresholds defined in this document: large-scale process unit loss, hydrocarbon release with fire or explosion potential, or loss of refined product supply to military logistics fuel contracts -- no autonomous Gen AI action is permissible. A human command broker must be in the control loop for every change to a physical process parameter that a Gen AI component recommends. ML algorithms that are deterministic and formally testable may operate autonomously at this tier because their behavior is bounded and verifiable; they do not require a command broker. At mid-level criticality, Gen AI informs operators, and operators decide; ML may operate autonomously. At lowest criticality, Gen AI may act as a virtual assistant with limited autonomous action; ML operates with full autonomous behavior. For the petroleum sector Design Basis, the highest-criticality tier governs all applications that control or influence process unit operations, fired heater management, safety instrumented system parameters, pressure relief logic, and emergency shutdown systems.

The Command Broker. The command broker is the human operator who decides -- who brokers -- a course of action based on process observables and Gen AI advisory output. The

command-broker architecture requires that the control system enforce the broker function mechanically: the system must be incapable of executing a physical process command originating from a Gen AI advisory component without a positive human confirmation step that is local, auditable, and not itself executable remotely or by another software component. ML components operating autonomously below the Bright Line do not require command-broker interposition; their deterministic, testable behavior is the architectural basis for autonomous operation. A vendor characterization of embedded Gen AI as providing "autonomous process optimization" or "direct control integration" describes a capability that is inconsistent with Design Basis compliance in the highest-criticality petroleum sector applications. The vendor characterization does not govern; the Design Basis requirement governs.

Vendor-Embedded ML and Gen AI. Vendor-embedded ML and Gen AI in petroleum sector control systems must be assessed against the Bright Line before deployment in any highest-criticality application. The assessment must confirm three things: that the Gen AI component has no pathway to physical command execution without human confirmation; that ML components are deterministic and formally testable and may therefore operate autonomously; and that Gen AI components have been subjected to red-teaming by parties independent of the vendor, addressing petroleum sector-specific attack scenarios including adversarial inputs designed to produce unsafe process unit commands, fired heater misoperation, or safety instrumented system bypass.

6. DBA-MA Threat Framework for the Petroleum Sector

6.1 Threat Actor Taxonomy

The DBA-MA framework defines three threat actor tiers. The Design Basis Threat is drawn from the upper two.

Tier 1 — State Actor. Nation-states with advanced persistent threat capability, precision strike capability, and the demonstrated willingness to employ combined cyber, kinetic, and influence operations against petroleum infrastructure. Russia has executed sustained airstrikes on the Ukrainian refinery and fuel storage infrastructure. Iran has employed ballistic missiles and UAS against Saudi Aramco facilities. China has documented pre-positioning activity in U.S. critical infrastructure networks. All three present credible Tier 1 threats to the U.S. petroleum sector infrastructure.

Tier 2 — Near-State Proxy. Paramilitary and hybrid warfare organizations operating under state direction. The Houthi maritime campaign demonstrated Tier 2 capability against energy-sector targets at sea. Applied to Gulf Coast marine terminals, offshore platforms, or refinery waterfront access points, the Houthis' operational model — combined UAS, anti-ship missile, and surface drone attack — is directly relevant.

Tier 3 — Sophisticated Non-State Actor. Criminal organizations and ideologically motivated actors. The Colonial Pipeline ransomware incident is the Tier 3 reference event for the petroleum

sector: significant consequences, but substantially less severe than a Tier 1 or Tier 2 attack against the same infrastructure would produce.

6.2 Attack Modality Taxonomy

Three primary attack modalities are relevant to the petroleum sector.

Kinetic Direct. Physical destruction of refinery process units, pipeline infrastructure, storage tanks, or marine terminal facilities using conventional explosives, precision-guided munitions, or UAS-delivered munitions. Refinery process units — particularly elevated distillation columns, fluid catalytic crackers, and fired heaters — are vulnerable to standoff kinetic attack in ways that many other industrial facilities are not.

Cyber-Kinetic Combined Arms. Integration of ICS/SCADA compromises with kinetic action. In the petroleum refining context, DCS compromise can manipulate process conditions — temperatures, pressures, flow rates, valve positions — to create process upsets that amplify the physical damage of a concurrent kinetic strike, suppress safety system response, and complicate post-attack restart. The pre-positioning phase may last months before kinetic execution.

Supply Chain and Logistics Denial. Interdiction of the supply chains for critical replacement equipment — refinery process unit components, pipeline segments, control system hardware — to extend recovery timelines after a kinetic attack. A Tier 1 adversary that understands heat exchanger and compressor replacement lead times (six to eighteen months) can design supply chain interdiction operations that sustain disruption long after the physical damage has been assessed, and repair has begun.

6.3 Bounding Logic

The Design Basis Threat for the petroleum sector is: Tier 1 cyber pre-positioning with Tier 2 kinetic execution; combined cyber-kinetic modality; target selection for maximum multi-product supply impact; timing at peak demand conditions; and simultaneity within a window that exceeds the sector's adaptive capacity. The concentration of Gulf Coast refineries, the limited pipeline interconnections to alternative refining regions in some product markets, and the long recovery timelines for major process unit damage make the petroleum sector particularly vulnerable to a coordinated, simultaneous attack on multiple facilities within the same regional cluster.

6.4 The Gulf Coast Concentration Vulnerability

The single most consequential structural vulnerability in the U.S. petroleum sector is the geographic concentration of refining capacity on the Gulf Coast. Approximately 45 percent of total U.S. refining capacity is located in a coastal corridor accessible via the Gulf of Mexico. This concentration reflects the economic logic of proximity to domestic crude production and

deep-water import terminals. Still, it creates a targeting opportunity that a sophisticated adversary will exploit: a small number of coordinated attacks against facilities within a single coastal corridor can produce a national-level supply disruption.

The simultaneity trap identified in the DBA-GAS sector framework applies equally here. The Gulf Coast refinery cluster's resilience mechanisms — inter-refinery crude and product sharing, pipeline routing flexibility, and SPR drawdown — are designed for sequential recovery from single-facility events. A coordinated attack on multiple refineries within the same regional cluster exhausts those mechanisms simultaneously.

7. Consequence Interaction and Cross-Sector Dependencies

7.1 Military Logistics Dependency

The petroleum sector's most consequential cross-sector dependency under military-action conditions is the military logistics fuel supply chain. U.S. military aircraft, ground vehicles, and naval vessels depend on jet fuel (JP-8/Jet-A), diesel, and marine distillate produced by domestic refineries and distributed through Defense Logistics Agency contracts. A sustained disruption of Gulf Coast refining capacity that reduces jet fuel and diesel production constrains military operational tempo — sortie rates, ground vehicle operational range, and naval logistics — at precisely the moment when military capacity is most needed.

Pre-positioned military fuel reserves and strategic fuel stockpiles provide a buffer against short-duration refinery outages. For disruptions extending beyond weeks, operational constraints begin to affect force readiness. This feedback loop — military action degrades the refining capacity that supports military operations — is the adversary's strategic rationale for targeting petroleum infrastructure in a conflict.

7.2 Transportation and Commerce Cascade

Refined product supply disruptions cascade through transportation and commerce, compounding the direct impact on supply. Trucking fuel price spikes and diesel rationing affect freight costs and delivery schedules across the entire supply chain. Aviation fuel shortfalls affect both commercial aviation and military airlift. Agricultural diesel shortfalls during planting or harvest seasons have implications for food security. The cascading consequences of a major refinery disruption reach sectors — agriculture, manufacturing, emergency services — that do not directly consume petroleum products but depend on the transport and logistics networks that do.

7.3 Environmental Consequence as a Strategic Amplifier

Environmental consequences of petroleum sector attacks — crude oil spills, product releases, refinery fire-suppression runoff — amplify the strategic effect of military action in ways that do

not apply to the nuclear or gas sectors. A crude oil spill from a sabotaged pipeline into a navigable waterway produces environmental damage, remediation costs, and regulatory responses that extend the effective disruption well beyond the physical repair timeline. An adversary who designs an attack to maximize environmental consequence — targeting facilities near sensitive waterways, coastal estuaries, or groundwater recharge zones — achieves a disruption multiplier through the regulatory and remediation burden rather than through physical damage alone.

This environmental-consequence amplification mechanism is unique to the petroleum sector among the sectors addressed in the DBA-MA framework. It is addressed in the facility-class documents for offshore production and marine terminals, where exposure to the aquatic environment is greatest.

7.4 Market Consequence and Price Signal Cascade

Petroleum markets respond to supply-disruption signals faster than physical supply chains do. A credible attack on Gulf Coast refining infrastructure — regardless of the actual supply impact — produces immediate price spikes in crude oil, gasoline, and diesel futures markets. Those price signals translate into retail fuel price increases within days, affecting consumer and commercial activity before any physical supply shortage develops. An adversary who understands this dynamic can design attacks to maximize the market signal — targeting high-visibility facilities rather than the most consequential facilities from a supply perspective — to achieve economic disruption through price effects that outrun the actual physical supply impact.

8. Protective Action and Recovery Architecture

8.1 Threat Detection Framework

Elevated threat indicators for the petroleum sector include: Intelligence Community reporting of adversary interest in petroleum refinery or terminal infrastructure; anomalous ICS network activity at refinery DCS systems including unusual process historian access or lateral movement between IT and OT network segments; physical surveillance of refinery fence lines or marine terminal access points from vessels or adjacent facilities; unusual acquisition patterns for commercially available explosives, pipeline mapping data, or process engineering documentation for specific refinery configurations; and anomalous crude or product procurement patterns that may indicate advance knowledge of supply disruption.

Imminent threat indicators include: confirmed physical intrusion at refinery or terminal perimeter; simultaneous DCS anomalies at multiple refineries within a single regional cluster; anomalous pressure readings at multiple pipeline nodes within a single system within a short time window; radar or visual contact with unidentified small surface vessels on approach vectors

to marine terminal infrastructure; and confirmed aerial UAS in restricted airspace over refinery or terminal facilities.

8.2 Decision Authority Framework

Facility operators retain authority to initiate emergency shutdown, process unit isolation, and site evacuation without external authorization. Sector coordination through the ONE-ISAC (formerly ONG-ISAC) and the Oil and Natural Gas Subsector Coordinating Council activates mutual aid and provides for TSA and CISA notification. Federal ESF-12 activation under DOE leadership coordinates federal support for supply restoration. SPR drawdown for a crude supply emergency requires the Secretary of Energy or Presidential authority, as appropriate. Emergency response under OPA 90 requires FOSC activation by the USCG or EPA as the designated Federal On-Scene Coordinator.

8.3 Federal Coordination Triggers

The following events constitute Federal Coordination Triggers -- thresholds that require mandatory federal coordination and, where authority exists, federal response action: confirmed coordinated attack on two or more petroleum refinery or terminal facilities within a single regional cluster within 48 hours; refined product supply disruption affecting military logistics fuel supply contracts above a defined threshold; ICS compromise with confirmed or strongly suspected nation-state attribution at two or more petroleum sector facilities; confirmed crude oil or product release to a navigable waterway resulting from circumstances consistent with adversary action; and confirmed fatalities at petroleum sector facilities in circumstances consistent with adversary action.

8.4 Recovery Sequencing Principles

Phase 1 — Immediate stabilization. Emergency shutdown of affected process units. Fire suppression activation. Pipeline isolation at nearest block valves. SPR drawdown authorization for crude supply emergency. Military fuel reserve activation for priority military consumers. Product pipeline emergency routing assessment.

Phase 2 — Short-term bridging. Undamaged refinery capacity maximization. Product import surge from Atlantic Basin refineries. SPR crude drawdown to supplement refinery feedstock. Demand reduction measures for civilian fuel supply. Military logistics adjustment to conserve priority fuel grades.

Phase 3 — Medium-term restoration. Refinery damage assessment and repair scope definition. Specialty contractor mobilization. Replacement equipment procurement. Sequential process unit restart following safety case re-validation and regulatory inspection. Supply chain monitoring for adversary interdiction of replacement equipment.

The local sovereignty principle applies to Phase 1 protective actions. Emergency shutdown systems, pressure relief devices, and fire and gas detection systems must operate passively, without external power, communications, or operator action. Phase 2 and Phase 3 actions require the deployment of external resources and are subject to the access-denial constraint under military-action conditions.

9. Facility Class Register and Inheritance Model

9.1 The Inheritance Model

Facility-class documents in the DBA-OIL series inherit the sector framework's threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards. They do not repeat this material. Each facility-class document adds what is specific to its asset profile: facility-class-specific bounding events, consequence chains, design parameters, and recovery architecture, which the sector framework establishes in general terms but does not develop for each asset class.

Each facility-class instrument must include a section explicitly identifying which elements are inherited from this document unchanged, which elements are inherited with facility-class-specific adaptation, and which elements are entirely locally defined. This inheritance map is a required component of every facility-class instrument at the time of first publication. It must be updated at each revision to reflect any changes to this sector framework.

Use cases issued under the sector framework apply the DBA-MA methodology to a specific attack vector or consequence chain within a facility class's scope. Policy companions outline the regulatory, legislative, or institutional actions required to address identified gaps. This modular architecture allows the series to grow as new facility classes are analyzed and new use cases are developed, without requiring revision of the foundational framework documents.

9.2 Facility Class Placement Rationale

The petroleum sector is organized around the physical pathway from wellhead through refining to distribution. Facility classes in the DBA-OIL series are defined by their position in that pathway and by the consequence chain their loss initiates. Petroleum refining (FC1) is placed first because refinery outage is the highest-consequence single-facility event in the domestic petroleum sector: the Gulf Coast refinery cluster processes approximately 45 percent of U.S. refining capacity, and coordinated action against it produces national fuel supply consequences within days. Offshore crude production (FC2) is placed second because wellhead outage reduces the feedstock available to the refinery system; its consequence chain runs upstream of refining and is amplified by the remote-access constraints of offshore operations. Terminals and storage (FC3) address the intermediate logistics layer that buffers production and distribution; their incapacitation severs the connection between production and consumer delivery regardless of

refinery status. Crude and product transport infrastructure (FC4) addresses the pipeline network that moves both crude feedstock to refineries and refined product to distribution points; its loss produces the supply chain disruption consequences that Colonial Pipeline demonstrated.

9.3 Facility Class Register

The following table is the current facility class register for the DBA-OIL series. Facility classes designated as Anticipated require specialist development by qualified petroleum sector engineers and safety analysts. Their inclusion in the register establishes the planned scope of the series; it does not represent completed analytical work.

Designation	Facility Class	NIPP Sector	Status	Primary Regulatory Anchor
DBA-OIL-FC1	Petroleum refining	Energy	Active -- v1.2	EPA RMP (40 CFR Part 68); OSHA PSM (29 CFR 1910.119); NERC CIP-014 (where applicable)
DBA-OIL-FC2	Offshore crude production -- platforms and associated pipeline	Energy	Active -- v0.1	BSEE 30 CFR Parts 250/254; USCG 33 CFR; PHMSA 49 CFR Part 195
DBA-OIL-FC3	Crude and product terminals and storage -- onshore and marine	Energy	Active -- v0.1	EPA SPCC (40 CFR Part 112); USCG 33 CFR; PHMSA 49 CFR Part 195; TSA Pipeline Security Directives
DBA-OIL-FC4	Crude and product transport -- trunk and product pipelines	Energy	Active -- v0.1	PHMSA 49 CFR Part 195; TSA Pipeline Security Directives; EPA SPCC

9.4 Development Sequence

FC2 (Offshore Crude Production) is the next facility-class instrument to be developed, driven by the maritime threat profile and the remote-access constraints that distinguish offshore operations from onshore refining. The Gulf of Mexico is the primary reference environment; the North Sea and Norwegian shelf are the primary allied-nation analogs. FC3 (Terminals and Storage) follows, addressing the tank farm fire, spill, and supply chain disruption consequence chains that are analytically distinct from both refining and transport. FC4 (Crude and Product Transport) follows FC3; Colonial Pipeline is the primary reference event, and the TSA mandatory cybersecurity directives provide the regulatory foundation on which the physical security design basis can be built.

10. Gaps and Open Items

10.1 Regulatory Gap: Absence of Mandatory Physical Security Standards

The petroleum sector lacks mandatory physical security standards equivalent to the electricity sector's NERC CIP-014. TSA's pipeline security guidelines are voluntary for physical security; TSA's mandatory cybersecurity directives following the Colonial Pipeline incident address only the cyber dimension. No regulatory mechanism compels the implementation of the Design Basis established in this document. Translating the Design Basis into enforceable standards — through a process modeled on the development of NERC CIP-014 — is the single most consequential action to close the gap this framework identifies.

10.2 Gulf Coast Concentration and Resilience Planning

The concentration of Gulf Coast refineries is the defining structural vulnerability of the U.S. petroleum sector. No federal program currently requires refiners, pipeline operators, or federal regulators to assess the system-level consequences of a simultaneous attack on the Gulf Coast cluster or to maintain the mutual aid, spare equipment, and alternative routing arrangements that would contain those consequences. Addressing this gap requires system-level consequence analysis at the federal planning level and pre-established coordination protocols among refiners, pipeline operators, DOE, and DOD.

10.3 SPR Positioning and Military-Action Scenarios

The Strategic Petroleum Reserve was designed for scenarios of international supply disruption, not domestic infrastructure attacks. Its geographic concentration on the Gulf Coast means that the most consequential domestic attack scenario — coordinated action against the Gulf Coast refinery cluster — may also degrade SPR drawdown and distribution capabilities. Federal emergency planning should assess whether the SPR's current geographic and operational posture is adequate for the military-action scenario consequence envelope, or whether distributed emergency reserves at non-Gulf-Coast locations are warranted.

10.4 Environmental Consequences and Military-Action Response

OPA 90 facility response plans address worst-case discharge scenarios under accidental conditions. They do not address the military-action scenario, in which the response resources called for by the facility response plan may be unable to access the site, the volume and rate of release may exceed the worst-case spill design basis, and the simultaneous occurrence of fires, explosions, and security threats creates a multi-hazard response environment that exceeds the planning basis of current environmental response frameworks. BSEE, EPA, and USCG should assess whether military-action scenario consequence analysis is needed as a supplement to existing OPA 90 planning requirements.

Revision History

Version	Date	Author	Description
1.0	April 2026	Tim Roxey	Initial version. Establishes the Design Basis and DBA-MA framework for the petroleum sector. Introduces the non-adversarial Design Basis methodology and bounding event categories. Defines the four characteristics of military action and their implications for the petroleum sector. Establishes sector-level threat taxonomy, consequence interaction framework, and facility-class instrument architecture. DBA-OIL-FC1 (Petroleum Refining) issued concurrently.
1.2	May 2026	Tim Roxey	Section 9 — "Facility Class Instruments" replaced with full "Facility Class Register and Inheritance Model" architecture (Sections 9.1–9.4). Existing inheritance model prose retained and extended with a mandatory inheritance-map requirement. New Sections 9.2 (Placement Rationale), 9.3 (five-column register table), and 9.4 (Development Sequence) added. Addresses Gap 2.2 of DBA Series Gaps and Recommendations v1.2.
1.3	June 2026	T. Roxey	Section 5.4 (The Command-Broker Architecture and the Bright Line) added. Establishes the three-tier criticality framework inherited from CB-Framework_Cross_Sector v0.1; defines Command Broker requirement for highest-criticality petroleum sector applications; establishes vendor-embedded ML and Gen AI assessment requirements. Section 8.3 renamed from "Bright Line Criteria" to "Federal Coordination Triggers"; body text updated accordingly. "Bright Line" is reserved exclusively for the ML/AI governance boundary throughout this document.
1.4	June 2026	T. Roxey	Housekeeping revision. "Brightline" corrected to "Bright Line" (two words) throughout §5.4 heading and body, consistent with series-wide convention. FC1 register entry updated from Active -- v1.0 to Active -- v1.2. v1.3 revision history entry added (previously undocumented). No substantive content changes.
1.5	June 2026	T. Roxey	Housekeeping revision. FC2 register entry updated from Anticipated to Active -- v0.1 (DBA-OIL-FC2: Offshore Crude Production). FC3 register entry updated from Anticipated to Active -- v0.1 (DBA-OIL-FC3: Crude and Product Terminals and Storage). No substantive content changes.
1.6	June 2026	T. Roxey	Housekeeping revision. FC4 register entry updated from Anticipated to Active -- v0.1 (DBA-OIL-FC4: Crude and Product Transport -- Trunk and Product Pipelines). All four DBA-OIL facility class instruments are now at first issue. No substantive content changes.